

Total Questions: 35

LEVEL – II

Set - B

Time: 1.5 hr.

Total Marks: 70

SECTION - 1

- This section contains **FIFTEEN (15)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 2

- This section contains **EIGHT (08)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each correct answer you will get **TWO** marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given.
- There are **NO NEGATIVE** marks for wrong answers.

SECTION - 3

- This section contains **NINE (09)** questions of **PASSAGE TYPE OR COMPREHENSION TYPE**.
- Under each paragraph there will be three questions that has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 4

- This section contains **THREE (03)** questions. These questions are **MATCHING TYPE BASED QUESTIONS**.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers

Section - 1

- A pinhole camera is used to view a 50 m tall building located 100 m away. If the length of the pinhole camera (distance from pinhole to screen) is 20 cm, what will be the height of the image formed on the screen?
A) 10 cm B) 1 cm
C) 20 cm D) 5 cm
- A family in Hyderabad uses several appliances, resulting in a total monthly consumption of 150 units for the month

of June. what is the total energy charge (excluding fixed charges and duty) that will appear on their bill?

0-50 Units: ₹1.95 per unit (for consumption up to 50 units/month).

51-100 Units: ₹3.10 per unit (for consumption up to 100 units/month).

101-200 Units: ₹4.80 per unit (for consumption 101-200 units/month)

A) ₹492.50

B) ₹465.00

C) ₹720.00

D) ₹1,500.00

3. Consider the following statements.

- I. In a physical change, a new substance is formed.
- II. Physical changes are usually reversible.
- III. In a chemical change, a new substance with different properties is formed.
- IV. Chemical changes do not involve any change in composition.

Select the correct combination of true (T) and false (F) statements.

	I	II	III	IV
A)	T	T	F	F
B)	F	T	F	T
C)	F	F	T	T
D)	F	F	F	T

4. A city is rapidly developing with more buildings, roads, and fewer open lands. After a few years, people observe that the groundwater level has decreased significantly. What is the main reason for this decrease?

- A) Increase in water seepage into the ground
- B) Decrease in effective area for seepage of water
- C) Increase in rainfall
- D) Reduction in population

5. A plant's leaves are turning pale yellow, especially between the veins, but the plant is receiving plenty of sunlight and water. This "Chlorosis" is often due to a deficiency of a specific mineral used to build the chlorophyll molecule. Which mineral is it?

- A) Iron
- B) Magnesium.
- C) Phosphorus.
- D) Potassium.

6. The stomach secretes Hydrochloric Acid (HCl) to kill bacteria and create an acidic

environment for protein digestion. However, the small intestine requires an alkaline (basic) environment for its enzymes to work. Which pair of organs provides the secretions necessary to neutralize the stomach acid as food enters the small intestine?

- A) Liver and Pancreas
- B) Salivary Glands and Esophagus.
- C) Large Intestine and Rectum.
- D) Kidneys and Bladder.

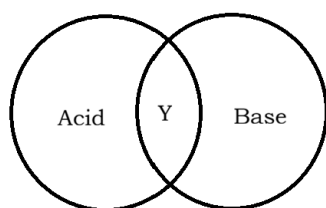
7. Blood viscosity (thickness) can change based on hydration. If a person is severely dehydrated, their blood becomes more viscous. How does this affect the "Circulatory System"?

- A) It makes the blood flow faster through the capillaries.
- B) It increases the resistance to flow, forcing the heart to work much harder to pump the blood.
- C) It prevents the blood from carrying urea to the kidneys.
- D) It causes the valves in the heart to stay open all the time.

8. Sun's heat reaches Earth through space. Which are incorrect?

- A) Radiation transfers heat in vacuum
- B) Conduction is not possible in vacuum
- C) Convection occurs in space
- D) Radiation travels at speed of light

9. A Venn diagram compares the properties of acids and bases.



Which of the following does not belong to Y region?

- Their aqueous solutions conduct electricity
 - They have pH values always below 7
 - They produce ions in aqueous medium
 - They change the colour of indicators
 - They turn red litmus blue
- A) (i), (ii) and (iii) only
 B) (ii) only
 C) (i), (iv) and (v) only
 D) (ii) and (v) only

10. Insects like grasshoppers have a "Tracheal System" for respiration. They do not have haemoglobin in their blood (which is why their blood is not red). How do their cells receive oxygen?

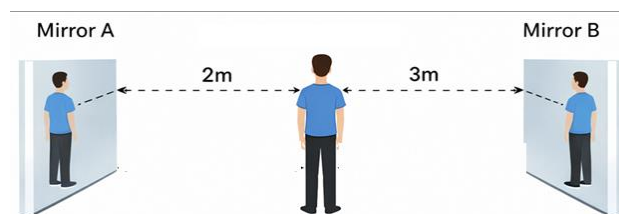
- A) Oxygen is absorbed through their legs and moves to the heart.
 B) A network of air tubes (tracheae) carries air directly from the spiracles to every cell in the body.
 C) They perform photosynthesis in their wings to make oxygen.
 D) They don't need oxygen and breathe nitrogen instead.

11. In "Double Fertilization," one male gamete fuses with the egg, and the other fuses with the secondary nucleus. If the egg cell of a plant has 12 chromosomes, how many chromosomes will be found in

the resulting "Endosperm" (the food storage tissue)?

- A) 12
 B) 24
 C) 36
 D) 48

12. What is the distance between the first image in Mirror A and the first image in Mirror B?



- A) 8 cm
 B) 5 cm
 C) 15 cm
 D) 10 cm

13. When 9 g of substance X is completely burnt, it produces 4 g of substance Y and the remaining mass is of substance Z.

Which of the following statements is correct?

- A) Mass of substance Z is 5 g
 B) Mass of substance Z is 13 g
 C) Total mass of products is less than the mass of reactant
 D) Mass of substance Y and Z together is greater than 9 g

14. Some seeds, like those of the "Xanthium," have hooks or spines, while seeds of "Coconuts" have a fibrous outer coat. Match these seeds to their respective dispersal agents:

- A) Xanthium — Wind; Coconut — Animals.
 B) Xanthium — Animals; Coconut — Water.
 C) Xanthium — Water; Coconut — Wind.
 D) Both are dispersed by Birds.

15. A group of researchers measured the "Breathing Rate" of four different animals: a Human, a Mouse, an Elephant, and a Cheetah. They found that Animal X had a breathing rate of over 100 breaths per minute, while Animal Y had a rate of only 6 breaths per minute. Identify Animal X and Y and the biological reason for this difference.

- A) X is the Elephant; Y is the Mouse. Large animals need more oxygen per second.
- B) X is the Mouse; Y is the Elephant. Smaller animals have higher metabolic rates and lose heat faster, requiring more oxygen for energy production.
- C) X is the Cheetah; Y is the Human. Fast animals always breathe faster even when sleeping.
- D) X is the Human; Y is the Cheetah. Humans have smaller lungs than all other mammals.

Section - 2

16. A student says: "If voltage is zero, current can still flow."

- A) Statement is incorrect
- B) Voltage is necessary for current
- C) Current can exist without charge
- D) Current depends on potential difference

17. A scientist performs the following experiments:

Substance P turns blue litmus red

Substance Q feels soapy and turns turmeric indicator red

Substance R reacts with baking soda to produce bubbles

Which of the following statements are correct?

- A) P is acidic in nature
- B) Q is a base
- C) R is a base because it produces bubbles
- D) R is acidic in nature

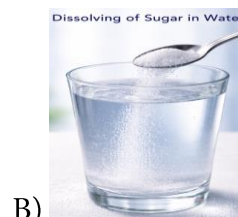
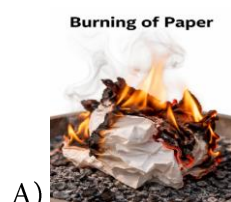
18. A student is studying a "Pitcher Plant" kept in a specialized glass chamber. The plant is provided with plenty of sunlight and water, but the air inside is filtered to remove all insects. Which of the following statements about this plant's nutritional status are correct?

- A) The plant will still be able to synthesize glucose because it possesses chlorophyll and performs photosynthesis.
- B) The plant will eventually show signs of nitrogen deficiency because it cannot supplement its nitrogen needs through insect digestion.
- C) The plant will turn into a complete saprotroph and start absorbing nutrients from the glass walls.
- D) The modified leaf (the pitcher) will continue to produce digestive enzymes even if no insect is trapped.

19. During the day, coastal areas experience wind blowing from sea to land. Which are correct?

- A) Land heats faster than sea
- B) Air above land rises
- C) Cool air from sea replaces hot air
- D) Process is due to convection

20. Which of the following are chemical changes?



Melting of Ice



Rusting of Iron



C)

D)

21. Ruminants, such as cows and deer, have adapted a unique way to digest cellulose. Which of the following are true about their nutrition?

- A) They have a four-chambered stomach, of which the "Rumen" is the first and largest.
- B) They chew the "Cud," which is partially digested food brought back from the rumen to the mouth.
- C) They possess special bacteria in their digestive tract that produce enzymes to break down cellulose.
- D) They swallow their food once and never chew it again, as the rumen does all the work.

22. The human heart is often called a "Double Pump." Which of the following statements correctly describe the pathway of blood in this system?

- A) Deoxygenated blood enters the Right Atrium and is pumped to the lungs by the Right Ventricle.
- B) The Pulmonary Vein is the only vein in the body that carries oxygenated blood from the lungs to the heart.
- C) The Aorta carries oxygenated blood at high pressure from the Left Ventricle to the rest of the body.
- D) Mixing of oxygenated and deoxygenated blood occurs in the ventricles to maintain a steady body temperature.

23. During a high-intensity 100-meter sprint, an athlete's body undergoes rapid

changes in respiration. Which of the following statements correctly describe these changes?

- A) The rate of breathing increases to provide more oxygen to the mitochondria for aerobic respiration.
- B) Muscle cells may switch to anaerobic respiration, producing Lactic Acid as a byproduct.
- C) Anaerobic respiration in the athlete's muscles produces ethanol and a large amount of energy.
- D) The accumulation of lactic acid is responsible for the muscle cramps experienced after heavy exercise.

Section - 3

COMPREHENSION-1

Water on Earth exists in three states—solid, liquid, and gas—and continuously changes from one form to another through the water cycle. Ice changes into liquid water by the process of (i), and water changes into vapour by (ii). Water vapour cools and changes back into liquid by (iii), forming clouds. When water falls back to Earth as rain or snow, the process is called (iv).

24. Which of the following correctly fills blanks (i) and (ii)?

- A) Freezing, Condensation
- B) Melting, Evaporation
- C) Evaporation, Melting
- D) Condensation, Freezing

25. Which option correctly represents the sequence of processes in blanks (ii), (iii), and (iv)?

- A) Evaporation → Condensation → Precipitation
- B) Condensation → Evaporation →

Precipitation

C) Precipitation → Evaporation → Condensation

D) Evaporation → Precipitation → Condensation

26. If the temperature decreases significantly, which process among the blanks is most likely to increase?

- A) (i) B) (ii)
C) (iii) D) (iv)

COMPREHENSION-2

Arjun studied a diagram of the human circulatory system and found that blood passes through the heart twice during one complete cycle. He observed that oxygenated and deoxygenated blood are kept separate.

27. Double circulation is absent in:

- A) Humans B) Birds
C) Fish D) Mammals

28. The separation of oxygenated and deoxygenated blood helps in:

- A) Reducing blood pressure
B) Efficient oxygen supply
C) Faster digestion
D) Reduced heart rate

29. Separation of oxygenated and deoxygenated blood occurs due to:

- A) Two-chambered heart
B) Three-chambered heart
C) Four-chambered heart
D) Absence of valves

COMPREHENSION-3

An autonomous rover moves 100 m east in 20 s, then 100 m north in 20 s, and finally returns to the starting point in 20 s along a straight path while its speed changes uniformly.

30. The magnitude of average velocity of the rover for the entire journey is:

- A) 5 m/s (B) 0 m/s
C) 2.5 m/s D) 10 m/s

31. During the return journey, which of the following is correct?

- A) Acceleration is zero because path is straight
B) Velocity is constant because direction is fixed
C) Acceleration is non-zero due to change in speed
D) Both velocity and acceleration are constant

32. At the instant when the rover starts returning, which quantity changes discontinuously?

- A) Speed B) Velocity
C) Acceleration D) Distance

Section - 4

33.

Column A	Column B
a) Self-pollination	P) Between different plants
b) Cross-pollination	Q) Same flower
c) Wind pollination	R) Light pollen
d) Insect pollination	S) Sticky pollen

- A) a-Q, b-R, c-P, d-S
B) a-P, b-Q, c-S, d-R
C) a-Q, b-P, c-R, d-S
D) a-S, b-P, c-Q, d-R

34.

COLUMN -A	COLUMN-B
A. A thick wire produces less heat than thin wire	P. Resistance $\propto$ length
B. High resistance element glows	Q. Heat $\propto I^2R$
C. Increasing length increases resistance	R. Resistance inversely proportional to area
D. Same current gives different heat in resistors	S. Energy conversion to heat increases with resistance

- A) A-R, B-S, C-P, D-Q
 B) A-Q, B-R, C-P, D-S
 C) A-R, B-Q, C-S, D-P
 D) A-P, B-S, C-R, D-Q

35.

Acid	Primary Source
(a) Maleic acid	 (P)
(b) Ascorbic acid	 (Q)
(c) Oxalic acid	 (R)
(d) Lactic acid	 (S)

- A) a-Q, b-S, c-P, d-R
 B) a-P, b-S, c-Q, d-R
 C) a-Q, b-P, c-S, d-R
 D) a-R, b-S, c-Q, d-P